X-Ray Heating

Code Documentation

Y. S. V. Aditya

Contents

1 Associated Files and Folders

This document corresponds to the following notebook:

`/Notebooks/XRay_Heating.ipynb`

We import reusables from the following source file:

`/src/utils.py`

Relevant figures are stored in:

`/Figures/XRay_Heating`

2 Objective

In this document and the corresponding Jupyter notebook, we attempt to numerically solve the heating of the universe due to X-ray photons emitted by High Mass X-ray Binaries (HMXBs). We follow the prescription adopted by [4] closely, with the aim of solving:

$$\frac{\mathrm{d}T_{\mathrm{K}}}{\mathrm{d}t} = -2HT_{\mathrm{K}} + \frac{T_{\mathrm{K}}}{\mu}\frac{\mathrm{d}\mu}{\mathrm{d}t} + \frac{2\mu m_{\mathrm{p}}}{3k_{\mathrm{B}}\rho_{\mathrm{b}}}(\mathcal{H} - \Lambda), \tag{2.1}$$

an ODE for the kinetic temperature T_{K} of the inter-galactic medium (IGM). We discuss this equation in further detail in the Theory section.

3 Theory

The thermal history of the IGM is poorly understood. This history reflects the duration of reionization as well as the nature of the ionizing sources. The IGM temperature from $z = 2 - 5$ is is not consistent with what is expected solely from complete hydrogen reionization, and requires substantial energy, potentially from photoionization of HeII. An alternate method to probe the universe after the cosmic dark age involves the use of the 21-cm hyperfine transition signal from neutral hydrogen. We expect the sky map of the 21-cm signal to carry additional information in the form of anisotropies.

X-ray photons can escape host galaxies easily to thermalize and weakly ionize the low-density IGM. Figure 1 shows the relative contribution of active galactic nuclei (AGN), high mass X-ray binaries (HMXBs), and low mass X-ray binaries (LMXBs) to the X-ray background in terms of the source emissivities in erg$\cdot$s$^{-1}\cdot$Mpc^{-3} in the $2 - 10$ keV energy band. We see that above $z \sim 6$, HMXBs dominate the X-ray contribution.

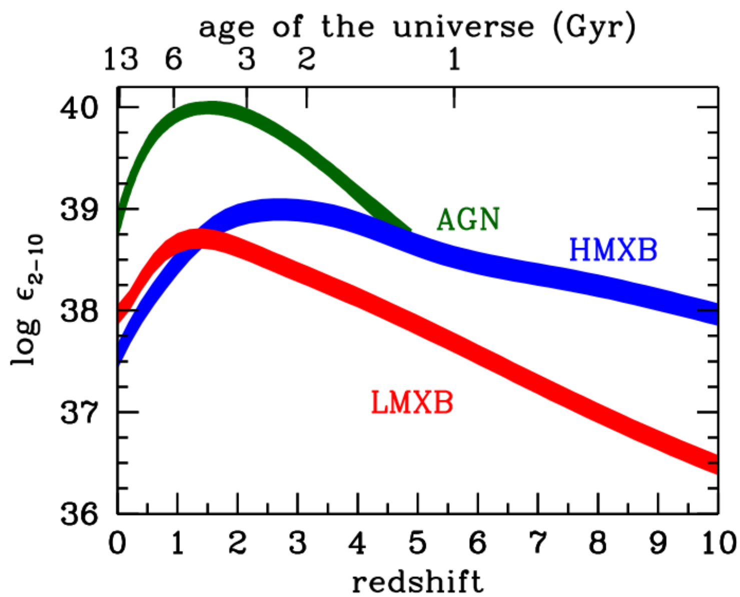

Figure 1: X-ray emissivities of Active Galactic Nuclei (AGN), high mass X-ray binaries (HMXBs) and low mass X-ray binaries (LMXBs) in the $2 - 10$ keV range as a function of redshift ([1], [4]).

The X-ray luminosity per unit star formation rate (SFR) from binaries is expected to be enhanced at higher redshifts owing to the metallicity Z.

3.1 Cosmic Star Formation History

For this work, we use two different models of comoving global star formation rate densities (SFRDs). First one is as given by [2]:

$$\dot{\rho}_* = \frac{(a + bz)h}{1 + (z/c)^d} \ \mathrm{M}_\odot\cdot\mathrm{yr}^{-1}\cdot\mathrm{cMpc}^{-3}, \tag{3.1}$$

where $a = 0.017$, $b = 0.13$, $c = 3.3$, and $d = 5.3$. cMpc denotes a comoving Megaparsec. The SFRD is normalized by the Salpeter IMF [5] between $0.1 - 100$ M$_\odot$. This is the primary SFRD model we use for most of our calculations.

The second SFRD model is given by [4] as:

$$\dot{\rho}_* = 0.01\frac{(1+z)^{2.6}}{1+[(1+z)/3.2]^{6.2}}\ \mathrm{M}_\odot\cdot\mathrm{yr}^{-1}\cdot\mathrm{cMpc}^{-3}. \tag{3.2}$$

The normalization factor of this SFRD model has been multiplied by 0.66 to convert from the Salpeter IMF to the Kroupa IMF [3].

Both the SFRD models have been plotted for comparizon in:

```
/Figures/XRay_Heating/SFRD_Models.png
```

3.2 Emissivity and Radiation Intensity

We begin by considering the total number of photons in an infinitesimal volume ΔV and a frequency range $\Delta\nu$ at a time t: $N = n_\nu \Delta V \Delta\nu$ where n_ν is the photon number density per unit frequency. After some time Δt, the number of photons must be conserved if they are neither absorbed nor created, but we have:

$$\Delta\nu(t+\Delta t) \approx \Delta\nu(t) + \Delta t\frac{\mathrm{d}\Delta\nu}{\mathrm{d}t}, \tag{3.3}$$

where $\mathrm{d}\nu/\mathrm{d}t = -\nu H(z)$, then we have:

$$\Delta\nu(t+\Delta t) \approx \Delta\nu(1 + H\Delta t). \tag{3.4}$$

Similarly for the number density n_ν, we find:

$$\frac{\mathrm{d}n_\nu}{\mathrm{d}t} = -2Hn_\nu \implies \mathrm{d}n_\nu = -2Hn_\nu\mathrm{d}t. \tag{3.5}$$

We note that n_ν is a function of both frequency and time; therefore, we have:

$$\frac{\partial n_\nu}{\partial \nu}\mathrm{d}\nu + \frac{\partial n_\nu}{\partial t}\mathrm{d}t = -2Hn_\nu\mathrm{d}t. \tag{3.6}$$

Then we have:

$$\frac{\partial n_\nu}{\partial t} = H\nu\frac{\partial n_\nu}{\partial \nu} - 2Hn_\nu. \tag{3.7}$$

Making the transformation to the specific intensity, defined as $J_\nu = (c/4\pi)h\nu n_\nu$, we get:

$$\frac{\partial J_\nu}{\partial t} - \nu H\frac{\partial J_\nu}{\partial \nu} + 3HJ_\nu = 0. \tag{3.8}$$

Now allowing for absorption and emission along the ray, we get from the standard radiative transfer equation:

$$\frac{\partial J_\nu}{\partial t} - \nu H\frac{\partial J_\nu}{\partial \nu} + 3HJ_\nu = -c\kappa_\nu J_\nu + \frac{c}{4\pi}\epsilon_\nu. \tag{3.9}$$

Here κ_ν is the absorption coefficient and ϵ_ν is the proper emissivity (in erg$\cdot$s$^{-1}\cdot$cm^{-3}). Transforming the second term, we get:

$$\frac{\mathrm{d}J_\nu}{\mathrm{d}t} = -(3H + c\kappa_\nu)J_\nu + \frac{c}{4\pi}\epsilon_\nu. \tag{3.10}$$

This is a first-order linear ODE, we define the integrating factor:

$$\mu(t) = \exp\left[\int_{t_0}^{t}(3H + c\kappa_{\nu_0})\mathrm{d}t'\right], \tag{3.11}$$

where all quantities with "0" subscript are initial quantities. Then:

$$\frac{\mathrm{d}(\mu J_\nu)}{\mathrm{d}t} = \mu \frac{c}{4\pi}\epsilon_{\nu_0} \implies \int_{\mu_0 J_{\nu_0}}^{\mu J_\nu} \mathrm{d}(\mu' J'_\nu) = \frac{c}{4\pi}\int_{t_0}^{t} \mu(t')\epsilon_\nu(t')\mathrm{d}t'. \tag{3.12}$$

Rearranging the equation:

$$J_\nu(t) = \frac{c}{4\pi}\int_{t_0}^{t} \frac{\mu(t')}{\mu(t)}\epsilon_\nu(t')\mathrm{d}t', \tag{3.13}$$

since $\mu(t)$ is independent of t'. Evaluating the ratio;

$$\frac{\mu(t')}{\mu(t)} = \frac{\exp\left[\int_{t_0}^{t'}(3H + c\kappa_{\nu_0})\mathrm{d}t''\right]}{\exp\left[\int_{t_0}^{t}(3H + c\kappa_{\nu_0})\mathrm{d}t''\right]} = \exp\left[-\int_{t'}^{t}(3H + c\kappa_{\nu_0})\mathrm{d}t''\right]. \tag{3.14}$$

We note here that:

$$\exp\left[-3H\int_{t'}^{t} H\mathrm{d}t''\right] = \exp\left[-3\int_{a'}^{a}\frac{1}{a''}\mathrm{d}a''\right] = \left(\frac{a'}{a}\right)^3. \tag{3.15}$$

Defining the quantity $\tau_{\text{eff}} = \int_{t'}^{t}\mathrm{d}t'' c\kappa_{\nu_0} = \int_{z'}^{z}\mathrm{d}z'' c\kappa_{\nu_0}|\mathrm{d}t''/\mathrm{d}z''|$, we get:

$$J_\nu(z) = \frac{c}{4\pi}\int_{z_0}^{z}\mathrm{d}z'\frac{\mathrm{d}t'}{\mathrm{d}z'}\epsilon_{\nu'}(z')\left(\frac{1+z}{1+z'}\right)^3 e^{-\tau(\nu_0,z,z')}. \tag{3.16}$$

Setting $t_0 = 0 \implies z_0 = \infty$:

$$J_\nu(z) = \frac{c}{4\pi}\int_{z}^{\infty}\frac{\mathrm{d}z'}{(1+z')H(z')}\epsilon_{\nu'}(z')\left(\frac{1+z}{1+z'}\right)^3 e^{-\tau(\nu_0,z,z')}, \tag{3.17}$$

where $\nu = \nu_0(1+z) = \nu'(1+z)/(1+z')$. Converting to comoving emissivity:

$$J_\nu(z) = \frac{c}{4\pi}\int_{z}^{\infty}\frac{\mathrm{d}z'}{(1+z')H(z')}\epsilon_{\nu'}(z')(1+z)^3 e^{-\tau}, \tag{3.18}$$

where $\epsilon_{\nu'}$ is now comoving. Now, we have $J_\nu\mathrm{d}\nu = J_E\mathrm{d}E \implies J_E = J_\nu/h$, and $\epsilon_\nu\mathrm{d}\nu = \epsilon_E\mathrm{d}E \implies \epsilon_E = \epsilon_\nu/h$. Therefore, we have:

$$J_E(z) = \frac{c}{4\pi}\int_{z}^{\infty}\frac{\mathrm{d}z'}{(1+z')H(z')}\epsilon_{E'}(z')(1+z)^3 e^{-\tau}, \tag{3.19}$$

where $E' = E_0(1+z') = E(1+z')/(1+z)$. Here τ can be written as:

$$\tau(E', z') = c\int_{z}^{z'}\frac{\mathrm{d}\overline{z}}{(1+\overline{z})H(\overline{z})\lambda_{\mathrm{X}}(\overline{E},\overline{z})}. \tag{3.20}$$

Here λ_{X} is the mean free path for photoelectric absorption of a photon of energy:

$$\overline{E} = E'\frac{(1+\overline{z})}{(1+z')}. \tag{3.21}$$

The mean free path is given as:

$$\lambda_{\mathrm{X}}(\overline{E},\overline{z}) = \frac{1}{\sum_i n_i\sigma_i}. \tag{3.22}$$

The effective photoelectric absorption cross-section in a neutral IGM can be approximated as $\sigma \equiv \sigma_{\mathrm{HI}} + y\sigma_{\mathrm{HeI}} \simeq (3.9\times10^{-23}\ \mathrm{cm}^2)(E/\mathrm{keV})^{-3.2}$ accurate to 82% within the photon energy range of $0.1 - 3$ keV [4], as we have shown in the figure:

/Figures/XRay_Heating/MFP_Ratios.png

Here y is the helium abundance by number, we can calculate it as:

$$N_{\mathrm{H}} = \frac{XM_{\mathrm{b}}}{m_{\mathrm{p}}}, N_{\mathrm{He}} = \frac{YM_{\mathrm{b}}}{4m_{\mathrm{p}}}, \tag{3.23}$$

where $X = 0.75$, $Y = 0.25$, N is the number of particles and M_{b} is the total baryonic mass. Therefore, the ratio of the number densities is given as:

$$y \equiv \frac{n_{\mathrm{He}}}{n_{\mathrm{H}}} = \frac{Y}{4X} \approx 0.083. \tag{3.24}$$

The photoelectric absorption cross-sections of individual species have been taken from [6]. The proper mean free path is then given as:

$$\lambda_{\mathrm{X}}(E, z) = (43.7 \text{ Mpc})(E/\text{keV})^{3.2} \left(\frac{1+z}{10}\right)^{-3}. \tag{3.25}$$

Now that we have derived the specific-intensity at redshift z, we calculate this quantity which will be required for the X-ray heating calculation that is to follow. We have followed two methods to do this, both of which vary in the method of finding the comoving emissivity ϵ_E, the first method is as prescribed in literature [4], and then our method (as we shall see, both the methods are effectively equivalent).

4 Code

4.1 Importing Packages and Reusable Functions

We import `numpy` for ease of operations on arrays, and `matplotlib.pyplot` for plotting:

```
import numpy as np
import matplotlib.pyplot as plt
```

To import `utils.py` containing all the reusable functions, we import `sys`, and specify the relative path directory:

```
import sys
sys.path.append('../src/')
```

Next we import `utils.py`. This has to be done every time any changes are maded to the `utils.py` file, and for ssafe measure, we refresh the import each time:

```
import utils
import importlib
importlib.reload(utils)
```

4.2 Cosmological Parameters

The Cosmological parameters that are used throughout this file are given in Table 1. We have assumed $\Omega_{\mathrm{r}} = 0$.

Parameter	Value
h	0.67
Ω_m	0.315
Ω_b	0.049
Ω_Λ	0.685
ρ_c	$(2.78 \times 10^{11})h^2\ \mathrm{M}_\odot{\cdot}\mathrm{Mpc}^{-3}$

Table 1: Cosmological parameters used in the code.

4.3 Scattering Cross Sections

The scattering cross sections are given by the fitting function [6]:

$$\sigma(E) = \sigma_0 F(y)\ \mathrm{Mb},\ x = \frac{E}{E_0} - y_0,\ y = \sqrt{x^2 + y_1^2}, \tag{4.1}$$

where:

$$F(y) = \left[(x-1)^2 + y_w^2\right] y^{0.5P-5.5} \left(1 + \sqrt{\frac{y}{y_a}}\right)^{-P}. \tag{4.2}$$

Here E_0, σ_0, y_a, P, y_w, y_0, and y_1 are fitting parameters.

References

[1] James Aird et al. *The evolution of the X-ray luminosity functions of unabsorbed and absorbed AGNs out to z 5*. 2015. arXiv: 1503.01120 [astro-ph.HE]. URL: https://arxiv.org/abs/1503.01120.

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